

Course: Cloud and Network Security CNS-C2-2026

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Week 1 Assignment 1:

Examine TCP/IP and OSI Models in Action

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## Objectives

Part 1: Examine HTTP Web Traffic

Part 2: Display Elements of the TCP/IP Protocol Suite

## Introduction

This activity establishes a foundation for understanding the TCP/IP protocol suite and its relationship to the OSI model, using Cisco Packet Tracer's Simulation mode to observe data as it crosses the network.

As data moves through the network, it is broken into smaller units called Protocol Data Units (PDUs), each tagged with its protocol and associated OSI/TCP-IP layer. Simulation mode allows these PDUs to be inspected layer by layer. This is done by requesting a webpage from a web server via the browser on a client PC, then tracking the PDU through each step of the exchange.

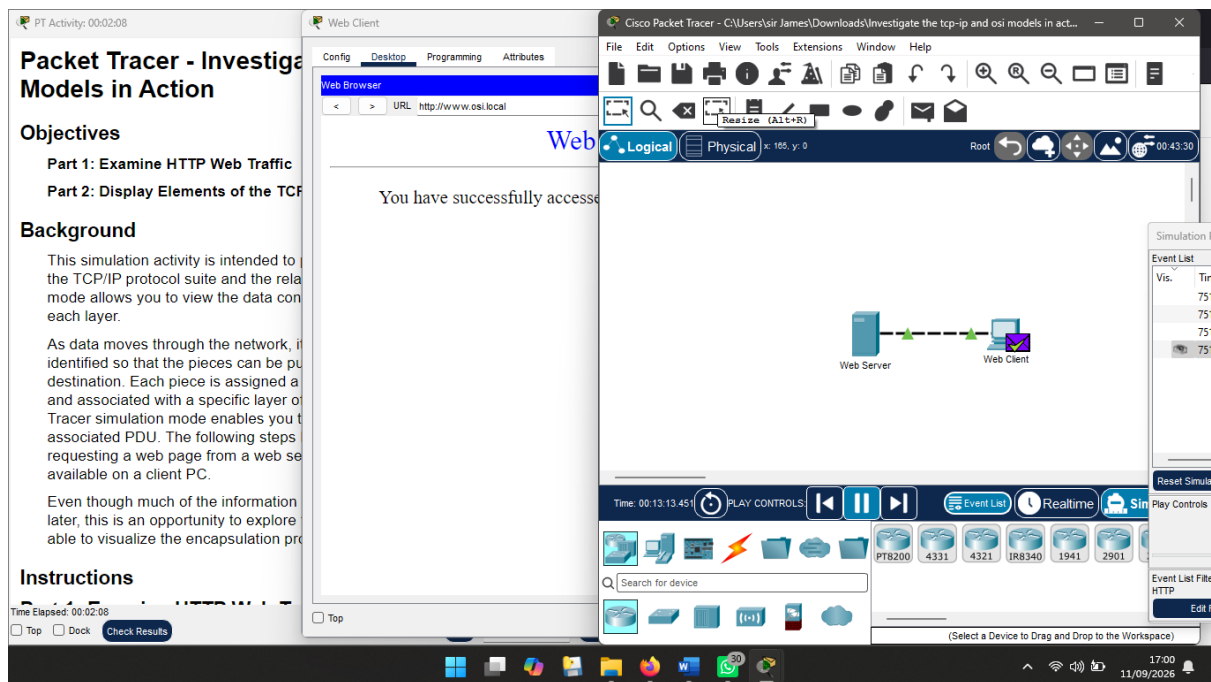
This activity's purpose is to build familiarity with Packet Tracer and visualize the encapsulation process.

## Part 1: Examine HTTP Web Traffic

### Switch from Realtime to Simulation mode

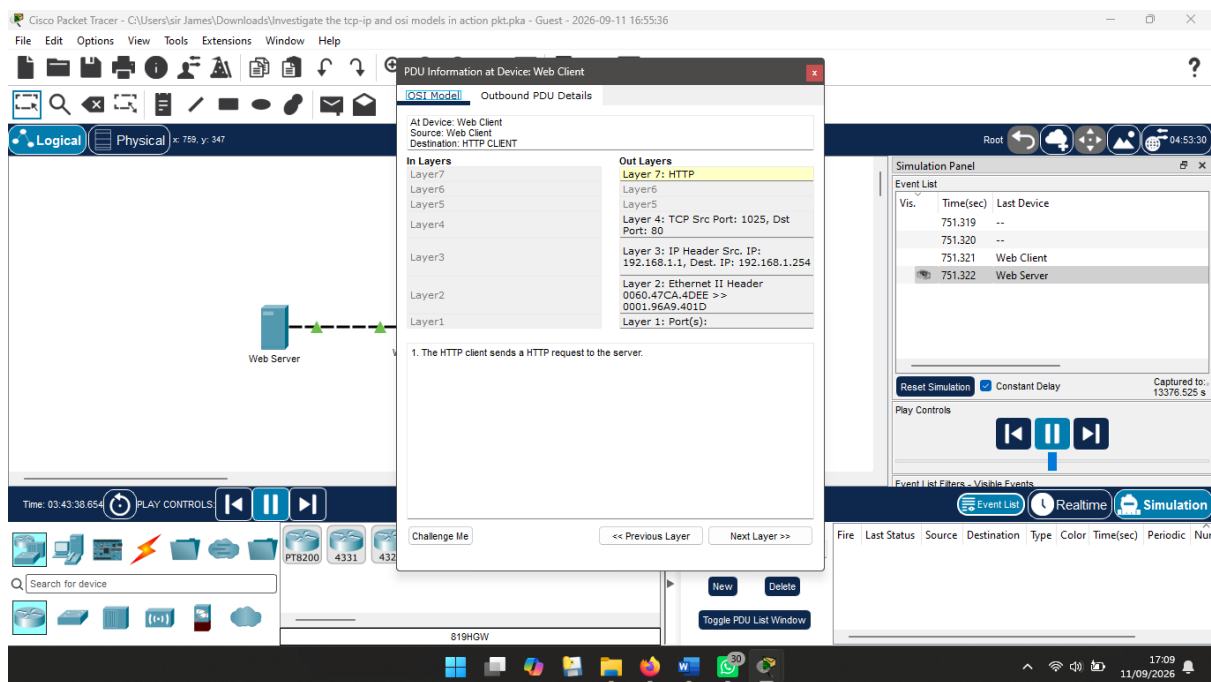
The screenshot shows the Cisco Packet Tracer interface. On the left, the 'Edit ACL Filters' dialog box is open, displaying a list of protocols under the 'Misc' tab. The 'HTTP' checkbox is selected. Below the list, there are buttons for 'Edit Filters' and 'Show All/None'. On the right, a web browser window is open, displaying a document titled 'Assignment-1-Packet-Tracer-Investi...'. The document contains instructions for the activity, including a section for 'Part 1: Examine HTTP Web Traffic' and 'Step 1: Switch from Realtime to Simulation mode'. The browser window also shows a 'Share' button and a page number 'Page 1 of 4'.

## Generate Web traffic



## Fist Event List

## Osi Model Details



Ensure that the OSI Model tab is selected.

Under the Out Layers column, click Layer 7.

### Questions:

What information is listed in the numbered steps directly below the In Layers and Out Layers boxes for Layer 7? HTTP

What is the Dst Port value for Layer 4 under the Out Layers column? 80

What is the Dest. IP value for Layer 3 under the Out Layers column? 192.168.1.254

What information is displayed at Layer 2 under the Out Layers column?

```
Layer 2: Ethernet II Header
0060.47CA.4DEE >>
0001.96A9.401D
```

### Outbound PDU Details

The screenshot displays the Cisco Packet Tracer interface with a network diagram showing a Web Server and a Web Client. The 'PDU Information at Device: Web Client' window is open, showing the 'Outbound PDU Details' tab. The PDU is an Ethernet II frame. The header fields are as follows:

| Ethernet II            |                |
|------------------------|----------------|
| Field                  | Value          |
| PREAMBLE               | 101010...10    |
| DEST ADDR              | 0001.96A9.401D |
| SRC ADDR               | 0060.47CA.4DEE |
| TYPE                   | 0800           |
| DATA (VARIABLE LENGTH) |                |
| FCS                    | 0x0000         |

The IP section shows:

| IP                     |               |
|------------------------|---------------|
| Field                  | Value         |
| VER                    | 4             |
| HL                     | 5             |
| DSCP                   | 0x00          |
| TL                     | 122           |
| ID                     | 0x0004        |
| FRAG OFFSET            | 0x000         |
| TTL                    | 128           |
| PRO                    | 0x06          |
| CHKSUM                 |               |
| SRC IP                 | 192.168.1.1   |
| DST IP                 | 192.168.1.254 |
| DATA (VARIABLE LENGTH) |               |

The TCP section shows:

| TCP                    |       |
|------------------------|-------|
| Field                  | Value |
| SOURCE PORT            | 1025  |
| DESTINATION PORT       | 80    |
| SEQUENCE NUMBER        | 1     |
| ACKNOWLEDGEMENT NUMBER | 1     |

### Questions:

What is the common information listed under the IP section of PDU Details as compared to the information listed under the OSI Model tab? With which layer is it associated?

The common info is the Source and destination Ips, both are listed in OSI model layer 3.

What is the common information listed under the TCP section of PDU Details, as compared to the information listed under the OSI Model tab, and with which layer is it associated?

The common info is the Source and destination ports; both are listed in OSI model layer 4.

What is the Host listed under the HTTP section of the PDU Details? What layer would this information be associated with under the OSI Model tab? The host: www.osi.local which will be associated with layer 7 under OSI model.

## Second Event List

PDU Details tab provides even more detailed information than is listed under Layer 2 on the OSI Model tab. The Outbound PDU Details provides more descriptive and detailed information. The values under DEST MAC and SRC MAC within the Ethernet II Header appear on the OSI Model tab under Layer 2 as such. Questions:

What is the common information listed under the In Layers column, as compared to the information listed under the Out Layers column, and with which layer is it associated?

What is the common information listed under the Out Layers column, as compared to the information listed under the In Layers column, and with which layer is it associated?

What is the Host listed under the HTTP section of the Out Layers column, and with which layer is it associated?

h. Click the next colored square box under the Event List. Only Layer 1 is active (not grayed out). The device from the buffer and placing it on to the network.

i. Advance to the next HTTP Type box within the Event List. Notice the direction of the arrow directly under the column, indicating the direction the data is travelling through these layers making note of the items previously viewed. At the top of the column the arrow points to the right. This now sending the information back to the client.

Comparing the information displayed in the In Layers column with that of the Out Layers column, what are the major differences?

Click the Inbound and Outbound PDU Details tab. Review the PDU Information at Device: Web Client.

Time Elapsed: 00:40:40

Top Dock Check Results Back 1/1 Next

Simulation Panel

| Vis.    | Time(sec) | Last Device |
|---------|-----------|-------------|
| 751.319 | --        | --          |
| 751.320 | --        | --          |
| 751.321 | --        | Web Client  |
| 751.322 | --        | Web Server  |

Reset Simulation Constant Delay Captured to: 58163.721 s

Play Controls

Event List Realtime Simulation

PT8200 4331 4321 1R8340 1941 2901 2911 815

2911

17:39 11/09/2026

## Third Event List

PDU Details tab provides even more detailed information than is listed under Layer 2 on the OSI Model tab. The Outbound PDU Details provides more descriptive and detailed information. The values under DEST MAC and SRC MAC within the Ethernet II Header appear on the OSI Model tab under Layer 2 as such. Questions:

What is the common information listed under the In Layers column, as compared to the information listed under the Out Layers column, and with which layer is it associated?

What is the common information listed under the Out Layers column, as compared to the information listed under the In Layers column, and with which layer is it associated?

What is the Host listed under the HTTP section of the Out Layers column, and with which layer is it associated?

h. Click the next colored square box under the Event List. Only Layer 1 is active (not grayed out). The device from the buffer and placing it on to the network.

i. Advance to the next HTTP Type box within the Event List. Notice the direction of the arrow directly under the column, indicating the direction the data is travelling through these layers making note of the items previously viewed. At the top of the column the arrow points to the right. This now sending the information back to the client.

Comparing the information displayed in the In Layers column with that of the Out Layers column, what are the major differences?

Click the Inbound and Outbound PDU Details tab. Review the PDU Information at Device: Web Server.

Time Elapsed: 00:41:25

Top Dock Check Results Back 1/1 Next

Simulation Panel

| Vis.    | Time(sec) | Last Device |
|---------|-----------|-------------|
| 751.319 | --        | --          |
| 751.320 | --        | --          |
| 751.321 | --        | Web Client  |
| 751.322 | --        | Web Server  |

Reset Simulation Constant Delay Capturing...

Play Controls

Event List Realtime Simulation

PT8200 4331 4321 1R8340 1941 2901 2911 815

2911

17:40 11/09/2026

This window contains both In Layers and Out Layers. Notice the direction of the arrow directly under the In Layers column; it is pointing upward, indicating the direction the data is travelling. Scroll through these layers making note of the items previously viewed. At the top of the column the arrow points to the right. This denotes that the server is now sending the information back to the client.

### Question:

Comparing the information displayed in the In Layers column with that of the Out Layers column, what are the major differences? The IP and MAC addresses swap, Ports flip too (Src 1025/Dst 80 becomes Src 80/Dst 1025) and in Layer 7 and layer 1, HTTP action changes from “receiving” to “sending”.

### Fourth event list

The screenshot displays the Cisco Packet Tracer interface. On the left, a document window titled 'PT Activity: 00:58:57' contains instructions for a lab activity. The instructions include steps n, i, j, and k, which involve clicking colored square boxes under the 'Event List' and 'Info' columns, and reviewing the 'Inbound and Outbound PDU Details' tab. Below these instructions, the document is divided into 'Part 2: Display Elements of the TCP/IP Protocol Suite' and 'Step 1: View Additional Events'. The main window shows the 'Cisco Packet Tracer' application. The 'PDU Information at Device: Web Client' window is open, displaying the 'OSI Model' and 'Inbound PDU Details' tabs. The 'OSI Model' tab shows the seven layers of the OSI model, with Layer 1 (Physical) and Layer 7 (Application) highlighted. The 'Inbound PDU Details' tab shows the details of the PDU being received by the Web Client. The details include: Source: Web Client, Destination: HTTP CLIENT, Layer 7: HTTP, Layer 6: Layer6, Layer 5: Layer5, Layer 4: TCP Src Port: 80, Dst Port: 1025, Layer 3: IP Header Src. IP: 192.168.1.254, Dest. IP: 192.168.1.1, Layer 2: Ethernet II Header 0001.9649.401D >> 0060.47CA.4DDE, and Layer 1: Port FastEthernet0. The 'Out Layers' column shows the layers of the PDU being sent back to the client, with Layer 1 (Physical) and Layer 7 (Application) highlighted. The 'Challenge Me' button is visible at the bottom of the PDU Information window. The bottom status bar shows 'Time Elapsed: 00:58:57', 'Top', 'Dock', 'Check Results', 'Back', '1/1', and 'Next'.

n. Click the next colored square box under the **Event List** > **Type** column. Only Layer 1 is active (not grayed out). The device is moving the frame from the buffer and placing it on to the network.

i. Advance to the next HTTP **Type** box within the **Event List** and click the colored square box. This window contains both **In Layers** and **Out Layers**. Notice the direction of the arrow directly under the **In Layers** column; it is pointing upward, indicating the direction the data is travelling. Scroll through these layers making note of the items previously viewed. At the top of the column the arrow points to the right. This denotes that the server is now sending the information back to the client.

Comparing the information displayed in the **In Layers** column with that of the **Out Layers** column, what are the major differences?

j. Click the **Inbound and Outbound PDU Details** tab. Review the PDU details.

k. Click the last colored square box under the **Info** column.

How many tabs are displayed with this event? Explain.

### Part 2: Display Elements of the TCP/IP Protocol Suite

In Part 2 of this activity, you will use the Packet Tracer Simulation mode to view and examine some of the other protocols comprising of TCP/IP suite.

#### Step 1: View Additional Events

a. Close any open PDU information windows.

b. In the **Event List Filters** > **Visible Events** section, click **Show All/None**.

What additional Event Types are displayed?

### Question:

How many tabs are displayed with this event? Explain.

Two tabs: OSI Model and Inbound PDU Details. Only two because this is the client receiving the final reply, nothing more gets sent back, so there's no "Outbound PDU Details" tab to show.

## Part 2: Display Elements of the TCP/IP Protocol Suite

### View Additional Events

PT Activity: 01:21:37

How many tabs are displayed with this event? Explain.

### Part 2: Display Elements of the TCP/IP Protocol Suite

In Part 2 of this activity, you will use the Packet Tracer Simulation and examine some of the other protocols comprising of TCP/IP.

#### Step 1: View Additional Events

- Close any open PDU information windows.
- In the **Event List Filters > Visible Events** section, click **Show All**.

What additional Event Types are displayed?

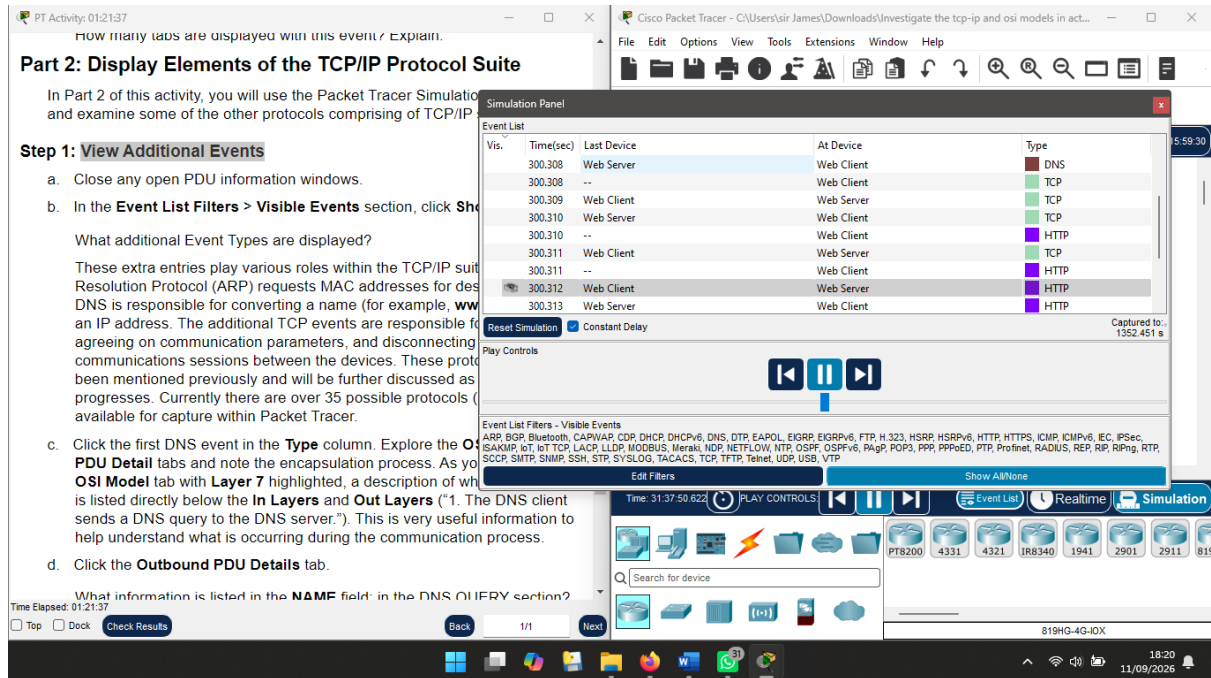
These extra entries play various roles within the TCP/IP suite. Address Resolution Protocol (ARP) requests MAC addresses for destination hosts. DNS is responsible for converting a name (for example, **www.osi.local**) to an IP address. The additional TCP events are responsible for agreeing on communication parameters, and disconnecting the communications sessions between the devices. These protocols have been mentioned previously and will be further discussed as the course progresses. Currently there are over 35 possible protocols (event types) available for capture within Packet Tracer.

- Click the first DNS event in the **Type** column. Explore the **OSI Model** and **PDU Detail** tabs and note the encapsulation process. As you look at the **OSI Model** tab with **Layer 7** highlighted, a description of what is listed directly below the **In Layers** and **Out Layers** ("1. The DNS client sends a DNS query to the DNS server."). This is very useful information to help understand what is occurring during the communication process.
- Click the **Outbound PDU Details** tab.

What information is listed in the **NAME** field: in the DNS QUERY section?

Time Elapsed: 01:21:37

☐ Top ☐ Dock   1/1



### Outbound PDU Details

PT Activity: 01:26:56

These extra entries play various roles within the TCP/IP suite. Address Resolution Protocol (ARP) requests MAC addresses for destination hosts. DNS is responsible for converting a name (for example, **www.osi.local**) to an IP address. The additional TCP events are responsible for agreeing on communication parameters, and disconnecting the communications sessions between the devices. These protocols have been mentioned previously and will be further discussed as the course progresses. Currently there are over 35 possible protocols (event types) available for capture within Packet Tracer.

- Click the first DNS event in the **Type** column. Explore the **OSI Model** and **PDU Detail** tabs and note the encapsulation process. As you look at the **OSI Model** tab with **Layer 7** highlighted, a description of what is occurring is listed directly below the **In Layers** and **Out Layers** ("1. The DNS client sends a DNS query to the DNS server."). This is very useful information to help understand what is occurring during the communication process.
- Click the **Outbound PDU Details** tab.

What information is listed in the **NAME** field: in the DNS QUERY section?

- Click the last DNS **Info** colored square box in the event list.

At which device was the PDU captured?

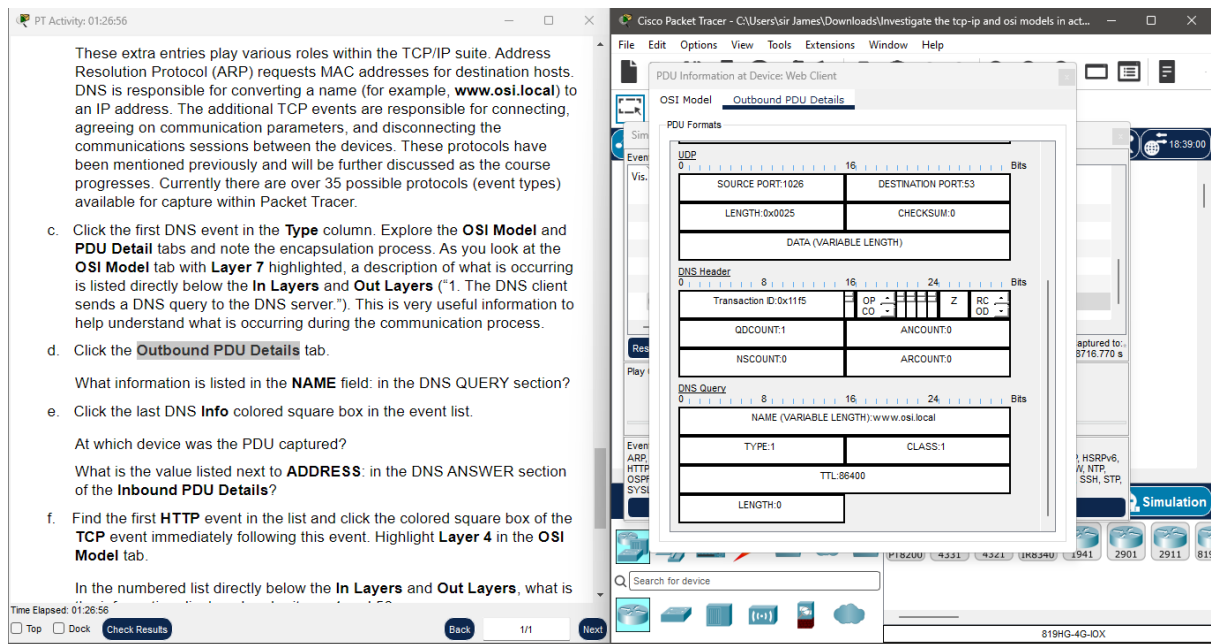
What is the value listed next to **ADDRESS**: in the DNS ANSWER section of the **Inbound PDU Details**?

- Find the first **HTTP** event in the list and click the colored square box of the **TCP** event immediately following this event. Highlight **Layer 4** in the **OSI Model** tab.

In the numbered list directly below the **In Layers** and **Out Layers**, what is

Time Elapsed: 01:26:56

☐ Top ☐ Dock   1/1



### Question:

What information is listed in the **NAME** field: in the DNS QUERY section? www.osi.local

## Last DNS Info coloured square box in the event list.

PT Activity: 01:28:47

DNS is responsible for converting a name (for example, **www.osi.local**) to an IP address. The additional TCP events are responsible for connecting, agreeing on communication parameters, and disconnecting the communications sessions between the devices. These protocols have been mentioned previously and will be further discussed as the course progresses. Currently there are over 35 possible protocols (event types) available for capture within Packet Tracer.

- Click the first DNS event in the **Type** column. Explore the **OSI Model** and **PDU Detail** tabs and note the encapsulation process. As you look at the **OSI Model** tab with **Layer 7** highlighted, a description of what is occurring is listed directly below the **In Layers** and **Out Layers** ("1. The DNS client sends a DNS query to the DNS server."). This is very useful information to help understand what is occurring during the communication process.
- Click the **Outbound PDU Details** tab.

What information is listed in the **NAME** field: in the DNS QUERY section?

- Click the last DNS **Info** colored square box in the event list.

At which device was the PDU captured?

What is the value listed next to **ADDRESS**: in the DNS ANSWER section of the **Inbound PDU Details**?

- Find the first **HTTP** event in the list and click the colored square box of the **TCP** event immediately following this event. Highlight **Layer 4** in the **OSI Model** tab.

In the numbered list directly below the **In Layers** and **Out Layers**, what is the information displayed under items 4 and 5?

TCP manages the connecting and disconnecting of the communications

Time Elapsed: 01:28:47

☐ Top ☐ Dock   1/1

Cisco Packet Tracer - C:\Users\sir James\Downloads\Investigate the tcp-ip and osi models in act...

File Edit Options View Tools Extensions Window Help

PDU Information at Device: Web Client

OSI Model Inbound PDU Details

At Device: Web Client  
Source: Web Client  
Destination: 192.168.1.254

| In Layers                                                        | Out Layers |
|------------------------------------------------------------------|------------|
| Layer 7: DNS                                                     | Layer 7    |
| Layer 6                                                          | Layer 6    |
| Layer 5                                                          | Layer 5    |
| Layer 4: UDP Src Port: 53, Dst Port: 1026                        | Layer 4    |
| Layer 3: IP Header Src. IP: 192.168.1.254, Dest. IP: 192.168.1.1 | Layer 3    |
| Layer 2: Ethernet II Header 0001.96A9.401D >> 0060.47CA.40EE     | Layer 2    |
| Layer 1: Port FastEthernet0                                      | Layer 1    |

1. FastEthernet0 receives the frame.

Event List

ARP, BGP, HTTP, HTTPS, OSPF, OSPF, SYSLOG, ...

Search for device

PT8200 4331 4321 1R8340 1941 2901 2911 815

819HG-4G-IOX

18:27 11/09/2026

### Question:

At which device was the PDU captured? Web client

What is the value listed next to **ADDRESS**: in the DNS ANSWER section of the Inbound PDU Details? Name: www.osi.local IP: 192.168.1.254.

## First HTTP event

PT Activity: 01:36:27

- Click the **Outbound PDU Details** tab.

What information is listed in the **NAME** field of the **Inbound PDU Details**?

- Click the last DNS **Info** colored square box in the event list.

At which device was the PDU captured?

What is the value listed next to **ADDRESS**: in the DNS ANSWER section of the **Inbound PDU Details**?

- Find the first **HTTP** event in the list and click the colored square box of the **TCP** event immediately following this event. Highlight **Layer 4** in the **OSI Model** tab.

In the numbered list directly below the **In Layers** and **Out Layers**, what is the information displayed under items 4 and 5?

TCP manages the connecting and disconnecting of the communications channel along with other responsibilities. the communication channel has been established.

- Click the last TCP event. Highlight Layer 4 in the steps listed directly below **In Layers** and **Out Layers**.

What is the purpose of this event, based on the last item in the list (should be item 4)?

### Challenge Questions

This simulation provided an example of a web server on a local area network (LAN). The client services running on the server. The server must respond for a client request. (Hint: Look at Layer 4 in the **OSI Model** tab for port information.)

Time Elapsed: 01:36:27

☐ Top ☐ Dock   1/1

Cisco Packet Tracer - C:\Users\sir James\Downloads\Investigate the tcp-ip and osi models in act...

File Edit Options View Tools Extensions Window Help

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

At Device: Web Client  
Source: Web Client  
Destination: HTTP CLIENT

| In Layers                                                        | Out Layers    |
|------------------------------------------------------------------|---------------|
| Layer 7                                                          | Layer 7: HTTP |
| Layer 6                                                          | Layer 6       |
| Layer 5                                                          | Layer 5       |
| Layer 4: TCP Src Port: 1026, Dst Port: 80                        | Layer 4       |
| Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254 | Layer 3       |
| Layer 2: Ethernet II Header 0001.96A9.401D >> 0060.47CA.40EE     | Layer 2       |
| Layer 1: Port(s):                                                | Layer 1       |

1. The HTTP client sends a HTTP request to the server.

Event List

ARP, BGP, HTTP, HTTPS, OSPF, OSPF, SYSLOG, ...

Search for device

PT8200 4331 4321 1R8340 1941 2901 2911 815

819HG-4G-IOX

18:35 11/09/2026

### Question:

In the numbered list directly below the In Layers and Out Layers, what is the information displayed under items 4 and 5?

- TCP processes payload data.
- TCP reassembles all data segments and passes to the upper layer.

### Last TCP event

The screenshot shows the Cisco Packet Tracer interface with a packet capture event selected. The 'OSI Model' tab is active, displaying the 'Inbound PDU Details' for a TCP segment. The 'In Layers' section shows Layer 4 (TCP) highlighted, with details: 'TCP Src Port: 1026, Dst Port: 80'. The 'Out Layers' section shows Layer 4 (TCP) highlighted, with details: 'TCP Seq: 1026, Dst Seq: 80'. The 'Challenge Questions' section contains the following text:

What is the value listed next to **ADDRESS** in the DNS **ANSWER** section of the **Inbound PDU Details**?

f. Find the first **HTTP** event in the list and click the colored square next to it. Highlight Layer 4 in the **OSI Model** tab.

In the numbered list directly below the **In Layers** and **Out Layers**, what is the information displayed under items 4 and 5?

TCP manages the connecting and disconnecting of the communication channel along with other responsibilities. This particular event indicates the communication channel has been **ESTABLISHED**.

g. Click the last **TCP** event. Highlight Layer 4 in the **OSI Model** tab. What is the purpose of this event, based on the information displayed under items 4 and 5?

**Challenge Questions**

This simulation provided an example of a web session between a client and a server on a local area network (LAN). The client makes requests for services running on the server. The server must be set up to listen for a client request. (Hint: Look at Layer 4 in the **OSI Model** information.)

Based on the information that was inspected during the Packet Tracer capture, what port number is the **Web Server** listening on for the web request?

What port is the **Web Server** listening on for a DNS request?

The bottom of the screenshot shows the 'Simulation' mode with a list of devices and their IP addresses: 192.168.1.1, 192.168.1.254, 192.168.1.253, 192.168.1.252, 192.168.1.251, 192.168.1.250, 192.168.1.249, 192.168.1.248, 192.168.1.247, 192.168.1.246, 192.168.1.245, 192.168.1.244, 192.168.1.243, 192.168.1.242, 192.168.1.241, 192.168.1.240, 192.168.1.239, 192.168.1.238, 192.168.1.237, 192.168.1.236, 192.168.1.235, 192.168.1.234, 192.168.1.233, 192.168.1.232, 192.168.1.231, 192.168.1.230, 192.168.1.229, 192.168.1.228, 192.168.1.227, 192.168.1.226, 192.168.1.225, 192.168.1.224, 192.168.1.223, 192.168.1.222, 192.168.1.221, 192.168.1.220, 192.168.1.219, 192.168.1.218, 192.168.1.217, 192.168.1.216, 192.168.1.215, 192.168.1.214, 192.168.1.213, 192.168.1.212, 192.168.1.211, 192.168.1.210, 192.168.1.209, 192.168.1.208, 192.168.1.207, 192.168.1.206, 192.168.1.205, 192.168.1.204, 192.168.1.203, 192.168.1.202, 192.168.1.201, 192.168.1.200, 192.168.1.199, 192.168.1.198, 192.168.1.197, 192.168.1.196, 192.168.1.195, 192.168.1.194, 192.168.1.193, 192.168.1.192, 192.168.1.191, 192.168.1.190, 192.168.1.189, 192.168.1.188, 192.168.1.187, 192.168.1.186, 192.168.1.185, 192.168.1.184, 192.168.1.183, 192.168.1.182, 192.168.1.181, 192.168.1.180, 192.168.1.179, 192.168.1.178, 192.168.1.177, 192.168.1.176, 192.168.1.175, 192.168.1.174, 192.168.1.173, 192.168.1.172, 192.168.1.171, 192.168.1.170, 192.168.1.169, 192.168.1.168, 192.168.1.167, 192.168.1.166, 192.168.1.165, 192.168.1.164, 192.168.1.163, 192.168.1.162, 192.168.1.161, 192.168.1.160, 192.168.1.159, 192.168.1.158, 192.168.1.157, 192.168.1.156, 192.168.1.155, 192.168.1.154, 192.168.1.153, 192.168.1.152, 192.168.1.151, 192.168.1.150, 192.168.1.149, 192.168.1.148, 192.168.1.147, 192.168.1.146, 192.168.1.145, 192.168.1.144, 192.168.1.143, 192.168.1.142, 192.168.1.141, 192.168.1.140, 192.168.1.139, 192.168.1.138, 192.168.1.137, 192.168.1.136, 192.168.1.135, 192.168.1.134, 192.168.1.133, 192.168.1.132, 192.168.1.131, 192.168.1.130, 192.168.1.129, 192.168.1.128, 192.168.1.127, 192.168.1.126, 192.168.1.125, 192.168.1.124, 192.168.1.123, 192.168.1.122, 192.168.1.121, 192.168.1.120, 192.168.1.119, 192.168.1.118, 192.168.1.117, 192.168.1.116, 192.168.1.115, 192.168.1.114, 192.168.1.113, 192.168.1.112, 192.168.1.111, 192.168.1.110, 192.168.1.109, 192.168.1.108, 192.168.1.107, 192.168.1.106, 192.168.1.105, 192.168.1.104, 192.168.1.103, 192.168.1.102, 192.168.1.101, 192.168.1.100, 192.168.1.99, 192.168.1.98, 192.168.1.97, 192.168.1.96, 192.168.1.95, 192.168.1.94, 192.168.1.93, 192.168.1.92, 192.168.1.91, 192.168.1.90, 192.168.1.89, 192.168.1.88, 192.168.1.87, 192.168.1.86, 192.168.1.85, 192.168.1.84, 192.168.1.83, 192.168.1.82, 192.168.1.81, 192.168.1.80, 192.168.1.79, 192.168.1.78, 192.168.1.77, 192.168.1.76, 192.168.1.75, 192.168.1.74, 192.168.1.73, 192.168.1.72, 192.168.1.71, 192.168.1.70, 192.168.1.69, 192.168.1.68, 192.168.1.67, 192.168.1.66, 192.168.1.65, 192.168.1.64, 192.168.1.63, 192.168.1.62, 192.168.1.61, 192.168.1.60, 192.168.1.59, 192.168.1.58, 192.168.1.57, 192.168.1.56, 192.168.1.55, 192.168.1.54, 192.168.1.53, 192.168.1.52, 192.168.1.51, 192.168.1.50, 192.168.1.49, 192.168.1.48, 192.168.1.47, 192.168.1.46, 192.168.1.45, 192.168.1.44, 192.168.1.43, 192.168.1.42, 192.168.1.41, 192.168.1.40, 192.168.1.39, 192.168.1.38, 192.168.1.37, 192.168.1.36, 192.168.1.35, 192.168.1.34, 192.168.1.33, 192.168.1.32, 192.168.1.31, 192.168.1.30, 192.168.1.29, 192.168.1.28, 192.168.1.27, 192.168.1.26, 192.168.1.25, 192.168.1.24, 192.168.1.23, 192.168.1.22, 192.168.1.21, 192.168.1.20, 192.168.1.19, 192.168.1.18, 192.168.1.17, 192.168.1.16, 192.168.1.15, 192.168.1.14, 192.168.1.13, 192.168.1.12, 192.168.1.11, 192.168.1.10, 192.168.1.9, 192.168.1.8, 192.168.1.7, 192.168.1.6, 192.168.1.5, 192.168.1.4, 192.168.1.3, 192.168.1.2, 192.168.1.1, 192.168.1.0.

### Question

What is the purpose of this event, based on the information provided in the last item in the list (should be item 4)?

The purpose of the event is to close the TCP connection — the device transitions its connection state to **CLOSED**, marking the formal end of the session between client and server now that all data has been successfully exchanged and acknowledged.

### Challenge answer

Based on the information that was inspected during the Packet Tracer capture, The Web Server listens on port 80 for HTTP (web) requests and port 53 for DNS requests.

### Conclusion

This activity demonstrated how the TCP/IP protocol suite and the OSI model work together to move data across a network. Using Packet Tracer's Simulation mode, the encapsulation and

de-encapsulation process was observed firsthand — from an HTTP request leaving the Web Client, through DNS resolution and TCP connection establishment, to the Web Server's response and eventual connection teardown. Examining the OSI Model and PDU Details tabs at each stage confirmed how data is broken into PDUs, tagged with addressing and port information at each layer, and reassembled at the destination. This exercise reinforced that while HTTP operates at Layer 7, its delivery depends on the coordinated function of every layer beneath it — TCP for reliable segment delivery, IP for logical addressing, and Ethernet for physical transmission — along with supporting protocols such as DNS and ARP. Overall, the activity provided practical insight into how theoretical networking models translate into real, observable network behaviour.